

Standard form for linear programming

$$\begin{aligned} \max_x \quad & C^T x \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0 \end{aligned}$$

$$Ax = b \Leftrightarrow \begin{aligned} Ax &\leq b \\ -Ax &\leq b \end{aligned}$$

add a slack variable per constraint

Equivalent form

$$\begin{aligned} \max_x \quad & C^T x \\ \text{s.t.} \quad & Ax \leq b \\ & x \geq 0 \end{aligned}$$

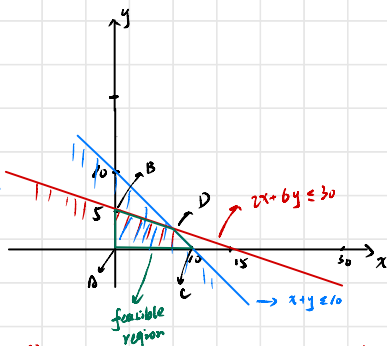
Example:

target = $20x + 50y$

price for A price for B profit

constraints:

$$\begin{aligned} x + y &\leq 10 && \text{time to prepare A and B are one hour each, total time} \leq 10 \\ 2x + 6y &\leq 30 && \text{oven time for A and B are 2 and 3 each, total time} \leq 30 \\ x &\geq 0 \\ y &\geq 0 \end{aligned}$$



To find the maximum, we only need to check

- Corner points
- A: Do nothing
 - B: Constrained by oven time
 - C: Constrained by prep time
 - D: Constrained by both
- here D is not an integer point
So need to check around.

why are these points enough?

definition

if $y, z \in S$ such that

$$x = \lambda y + (1-\lambda)z, \Rightarrow x = y = z$$

proof:

Because the feasible region S is convex,

any point x inside S can be expressed as a

sum of its corner points $v_1, \dots, v_k$

then $x = \lambda_1 v_1 + \dots + \lambda_k v_k$

where $\lambda_1 + \dots + \lambda_k = 1$

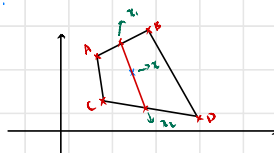
target function

$$\begin{aligned} \text{So } f(x) &= f(\lambda_1 v_1 + \dots + \lambda_k v_k) \\ &= \lambda_1 f(v_1) + \dots + \lambda_k f(v_k) \\ &\leq (\lambda_1 + \dots + \lambda_k) \max\{f(v_1), \dots, f(v_k)\} \\ &= \max\{f(v_1), \dots, f(v_k)\} \end{aligned}$$

this is the key argument for the proof and need to be proved itself

Why only inside point x can be expressed as a weighted sum of corner points?

Example:



$$\begin{aligned} x &= \lambda_1 x_1 + \lambda_2 x_2, \text{ where } \lambda_1 + \lambda_2 = 1 \\ x_1 &= \lambda_3 A + \lambda_4 B & \lambda_3 + \lambda_4 &= 1 \\ x_2 &= \lambda_5 C + \lambda_6 D & \lambda_5 + \lambda_6 &= 1 \\ \Rightarrow x &= \lambda_1 \lambda_3 A + \lambda_1 \lambda_4 B \\ &\quad + \lambda_2 \lambda_5 C + \lambda_2 \lambda_6 D \\ \text{where } \lambda_1 \lambda_3 + \lambda_1 \lambda_4 + \lambda_2 \lambda_5 + \lambda_2 \lambda_6 &= \lambda_1 (\lambda_3 + \lambda_4) + \lambda_2 (\lambda_5 + \lambda_6) \\ &= \lambda_1 + \lambda_2 \\ &= 1 \end{aligned}$$

formal proof uses the same idea by

induction on dimension

Analyze corner points

If we have n variables with m constraints and the non-negativity constraints.
(equation constraints as in standard form)

if $m=1 \Rightarrow$ We have n corner points $\rightarrow$ (n+1) is consider $x_1 = \dots = x_n = 0$

$$x_1 \neq 0, \quad x_2, \dots, x_n = 0$$

$$x_2 \neq 0, \quad x_1, x_3, \dots, x_n = 0$$

$\vdots$

$$x_n \neq 0, \quad x_1, \dots, x_{n-1} = 0$$

if $m > 1 \Rightarrow$ We have at most $\binom{m}{n}$ corner points.

$\downarrow$
this is only an upper bound. because not all selected m variables will have positive results.
 $\rightarrow$ We select m variables from all n variables to satisfy all the m equation constraints while all other $n-m$ variables $= 0$

Denote $A = (a_1, a_2, \dots, a_n)$. We know we need to satisfy the constraints

$$Ax = b$$

$\rightarrow$ if $b \in \text{Col}(a_1, \dots, a_n) \Rightarrow$ one solution or infinite solutions

if $b \notin \text{Col}(a_1, \dots, a_n) \Rightarrow$ there is no solution for the linear programming problem

The cases when there is only one solution or no solution are straightforward.
So we focus on the case when there is infinite solutions. Notice that

$$Ax = Bx_B + Nx_N = b$$

$\rightarrow$ ranging unselected $n-m$ columns and variables

we select m variables from all n .

Say $x_{i_1}, \dots, x_{i_m}$

$\rightarrow$ total possibility $\binom{m}{n}$

$(a_{i_1}, \dots, a_{i_m})$

$\rightarrow$ corresponding columns

Not all selected combinations have valid corresponding corner points

if B is invertible $\Rightarrow x_B = B^{-1}b$ and $x_N = 0$ is a valid corner points $\rightarrow$ Case (1)

if B is not invertible $\Rightarrow$ no solution $\rightarrow$ if these happen: no corner points for the selected m variables
 $\Rightarrow$ infinite solution $\rightarrow$
 $\Rightarrow$ 1 solution $\rightarrow x_B = \text{solution}$ and $x_N = 0$ is a valid corner points $\rightarrow$ Case (2)

Note: 1. x_N has to be 0, otherwise this is not corner point.

2. The $\binom{m}{n}$ combinations include all possible corner points.

$\downarrow$
if select more than m to satisfy the m constraints the corresponding columns can not be independent
So not inner points

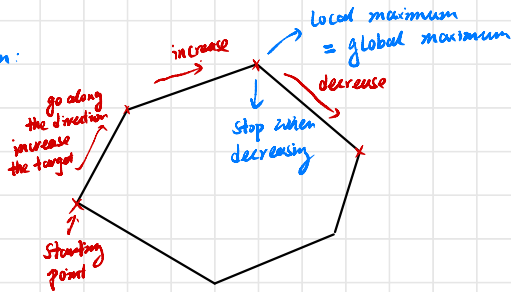
$\rightarrow$ if select less than m variables to satisfy the m constraints, then included in Case (2)

Solving a linear programming problem

We know an optimum (if exists) sits at a corner point.

However there are too many corner points $\binom{n}{n}$ to check...

Simplex algorithm:



Step 1. introduce slack variables if there are inequality constraints

Step 2. find a starting corner point

↳ the selected m variables: basis

non-selected $n-m$ variables: non-basis

→ this is harder than it sounds like.
need to solve $Ax=b$ and find
one solution where all $x \geq 0$.
easy to find one solution, but no
guarantee all $x \geq 0$

Step 3. if we cannot increase any non-basis variable
to make the target larger, this is
the maximum point, stop

↳ it can only increase
otherwise it will be negative

Step 4. choose a variable in non-basis (normally with the biggest coefficient)
that can increase the target when it increases, increase it from 0
until one basis variable hits 0 first.

↙ change basis: include the
selected one, drop the first
basis variable hit 0

↳ one of the m selected
variable

Step 5. Repeat Step 3 and 4

Example:

$$\max z = 3x_1 + 2x_2$$

$$\text{s.t. } x_1 + x_2 \leq 4$$

$$x_1 + 3x_2 \leq 6$$

$$x_1, x_2 \geq 0$$

Step 1, introduce slackness variables

the problem is equivalent to

$$\max z = 3x_1 + 2x_2$$

$$\text{s.t. } x_1 + x_2 + s_1 = 4$$

$$x_1 + 3x_2 + s_2 = 6$$

$$x_1, x_2, s_1, s_2 \geq 0$$

Step 2. choose an entry point

This example has an easy one

basis variables $B = \{s_1, s_2\} \rightarrow$ solve the equation $s_1 = 4, s_2 = 6$

non-basis variables $N = \{x_1, x_2\} \rightarrow$ set to 0, $x_1 = x_2 = 0$

So our entry point is $(0, 0, 4, 6) \rightarrow z = 0$

Step 3: note that $z = 3x_1 + 2x_2 \rightarrow$ express z as non-basis variables
 $\rightarrow 3$ is positive and > 2

We now change the basis to add x_1 in, while keeping $x_2 = 0$

$$\Rightarrow s_1 = 4 - x_1$$

$$\Rightarrow \text{when } x_1 = 4, s_1 = 0$$

$$s_2 = 6 - x_1$$

new basis: $x_1 = 4, s_2 = 2$

new non-basis: $x_2 = s_1 = 0$

$\rightarrow$ drop s_1 from basis

$$\rightarrow z = 12$$

this is always possible.

$$\text{Since } x_B = B^{-1}(b - N x_N)$$

$$\Rightarrow z = C_B^T x_B + C_N^T x_N$$

$\rightarrow$ express z in terms of non-basis variables

Step 4: Since

$$x_1 + x_2 + s_1 = 4$$

$$x_1 + 3x_2 + s_2 = 6$$

$$\Rightarrow x_1 = 4 - s_1 - x_2 \Rightarrow z = 3x_1 + 2x_2 = 12 - 3s_1 - x_2$$

both negative
stop here